

NAB2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NAB2 gene fusions from the msk_impact_50k_2026 study, covering 72 fusion events. 25/72 fusions (34.7%) were in-frame. 49/72 fusions (68.1%) retained the PF04905 domain. Domain retention was tested with Fisher's exact test ($p=0.11258$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for NAB2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 72 analyzed fusion events, identifying 14 distinct fusion partner genes. The most frequent partner was STAT6, observed in 53 events.

Domain retention

PF04905 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 20/25 (80.0%) of in-frame fusions retained the domain, $p=0.11258$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 1/25 (4.0%) of in-frame fusions disrupted the domain, $p=0.998249$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

Cutpoint detection

Cutpoint detection scanned 71 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 372 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is 6 aa from the nearest configured domain boundary at 366 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=49$) and "not_retained" ($n=22$). For Frame_status, retained: 20/21 (95.2%, 95% CI 77.3%-99.2%); not_retained: 5/14 (35.7%, 95% CI 16.3%-61.2%). Welch's t-test comparing Tumor_variant_count between groups gave means of 56.8776 vs 78.2727, $p=0.354421$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (14 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: NAB2 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
STAT6	53	0.7361111111111112	0.20043213737826426	0.0	0.34366576819407013	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3891339296891707	1
CUX2	1	0.013888888888888888	0.20043213737826426	0.0	0.948051948051948	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2631251900011857	2
PDZRN4	1	0.013888888888888888	0.20043213737826426	0.0	0.948051948051948	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2631251900011857	3
MGAT4C	2	0.027777777777777776	0.20043213737826426	0.0	0.9090909090909091	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26144770082369656	4
KRT6A	1	0.013888888888888888	0.20043213737826426	0.0	0.935064935064935	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2611771380531338	5
LRP1	3	0.041666666666666664	0.20043213737826426	0.0	0.8658008658008658	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2591208609968567	6
TMEM117	2	0.027777777777777776	0.20043213737826426	0.0	0.6558441558441559	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2234606878366836	7
KIF5A	1	0.013888888888888888	0.20043213737826426	0.0	0.6558441558441559	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2192940211700169	8
FMN1	3	0.041666666666666664	0.20043213737826426	0.0	0.23376623376623373	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16431566619166194	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
DLGAP2	1	0.0138888888888888	0.20043213737826426	0.0	0.23376623376623373	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.15598233285832858	10
FAM186A	1	0.0138888888888888	0.20043213737826426	0.0	0.23376623376623373	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.15598233285832858	11
LFNG	1	0.0138888888888888	0.20043213737826426	0.0	None	0.6664269678216079	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.14225576210987473	12
LGR5	1	0.0138888888888888	0.20043213737826426	0.0	0.09740259740259738	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.13552778740378313	13
MTOR	1	0.0138888888888888	0.20043213737826426	0.0	0.0	0.6664269678216079	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12091739779339353	14

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
319	0	49	4	18	0.0	0.00752854724253449	2.123288820003888
352	0	49	5	17	0.0	0.0020225947815764307	2.6940911176014084
364	0	49	13	9	0.0	8.560748975173139e-09	8.067488237477606
372	0	49	14	8	0.0	1.3283920823544527e-09	8.876673721601218
381	3	46	14	8	0.037267080745341616	5.884988789596151e-07	6.230254360113864
382	3	46	15	7	0.030434782608695653	1.0398205008988075e-07	6.983041624352472
400	4	45	15	7	0.04148148148148148	4.330878206429079e-07	6.36342402938942
426	4	45	16	6	0.03333333333333333	7.233726445978198e-08	7.140637918749475
434	5	44	16	6	0.04261363636363636	2.7039639334463545e-07	6.567999105485465
490	45	4	20	2	1.125	1.0	-0.0
502	46	3	20	2	1.5333333333333334	0.6419888956213136	0.19247248378799386
508	47	2	20	2	2.35	0.5828392348978783	0.23445122058408072
510	48	1	20	2	4.8	0.2249846907532149	0.6478470327668787

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
511	48	1	21	1	2.2857142857142856	0.5267605633802818	0.2783867465185951
521	49	0	21	1	None	0.3098591549295775	0.508835667896869

domain_disruption tables

frame_domain_contingency_table

1	12
24	34

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf04904	NAB conserved region 1 (NCD1) SAM domain	71	58	0	13	0.0

domain_retention tables

frame_domain_contingency_table

20	29
5	17

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	NAB conserved region 2 (NCD2)	71	49	13	9	0.3551986935220468

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	58	72	0.8055555555555556

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CUX2	1
DLGAP2	1
FAM186A	1
FMN1	3
KIF5A	1
KRT6A	1
LFNG	1
LGR5	1
LRP1	3
MGAT4C	2
MTOR	1
PDZRN4	1
STAT6	53
TMEM117	2

Per-event results (results.tsv)

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner_ gen e	fra me _sta tus	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost _do mai ns	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0052 994-T01-IM 6-1	P-0052 994-T0 1-IM6		FAM1 86A- NAB2	FAM 186 A	out- of-fr ame	thr ee_ _pr ime	574883 61	7	490	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2); NAB cons erve d re gion 1 (N CD1) SA M d oma in		FAM186A (NM_001145475) - NAB2 (NM_005967) rearrangement: c.193-1572 6:FAM186A_c.1469-34:NA B2inv Protein Fusion: out of frame {FAM186A:NAB2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015 989-T01-IM 6-2	P-0015 989-T0 1-IM6		FMN 1-NA B2	FMN 1	out- of-fr ame	thr ee _pr ime	574879 44	7	490	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2); NAB cons erve d re gion 1 (N CD1) SA M d oma in		FMN1 (NM_001103184) - NAB2 (NM_005967) Rearrangement : t(12,15)(q 13.3;q13.3)(chr12:g.57487 944::chr15:g.33270869) Protein Fusion: out of frame {FMN1:NAB2}	NA
EVT-P-0015 989-T05-IM 6-3	P-0015 989-T0 5-IM6		FMN 1-NA B2	FMN 1	out- of-fr ame	thr ee _pr ime	574879 44	7	490	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2); NAB cons erve d re gion 1 (N CD1) SA M d oma in		FMN1 (NM_001103184) - NAB2 (NM_005967) rearrangement: t(12,15)(q1 3.3;q13.3)(chr12:g.574879 44::chr15:g.33270870) Protein Fusion: out of frame {FMN1:NAB2}	NA
EVT-P-0015 989-T02-IM 6-4	P-0015 989-T0 2-IM6		FMN 1-NA B2	FMN 1	out- of-fr ame	thr ee _pr ime	574879 45	7	490	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2); NAB cons erve d re gion 1 (N CD1) SA M d oma in		FMN1 (NM_001103184) - NAB2 (NM_005967) Rearrangement : t(12,15)(q 13.3;q13.3)(chr12:g.57487 944::chr15:g.33270869) Protein Fusion: out of frame {FMN1:NAB2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020 678-T01-IM 6-6	P-0020 678-T01-IM6		KRT6A-NAB2	KRT6A	in-frame	three_prime	57486817	5	382	True	lost	0.0	False		NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain		KRT6A (NM_005554) - NAB2 (NM_005967) rearrangement: c.1204-574:KRT6A_c.1144-29:NAB2inv Protein Fusion: in frame {KRT6A:NAB2}	NA
EVT-P-0045 531-T01-IM 6-7	P-0045 531-T01-IM6		LGR5-NAB2	LGR5	unknown	three_prime	57488458	7	511	False	lost	0.0	False		NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain		LGR5 (NM_003667) - NAB2 (NM_005967) rearrangement: c.285-7692:LGR5_c.1532:NAB2dup Protein Fusion: mid-exon {LGR5:NAB2}	NA
EVT-P-0068 029-T01-IM 7-9	P-0068 029-T01-IM7		LRP1-NAB2	LRP1	unknown	three_prime	57487182	6	426	True	lost	0.0	False		NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain		LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.609:LRP1_c.1277-8:NAB2dup Protein Fusion: mid-exon {LRP1:NAB2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058 847-T01-IM 7-10	P-0058 847-T01-IM7		LRP1-NAB2	LRP1	unknown	three_prime	57486722	4	372	False	lost	0.0	False		NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain		LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.5662+685:LRP1_c.1114:NAB2dup Protein Fusion: mid-exon {LRP1:NAB2}	NA
EVT-P-0008 384-T01-IM 5-12	P-0008 384-T01-IM5		NAB2-CUX2	CUX2	out-of-frame	three_prime	57486406	4	364	True	disrupted	0.017964071856287425	True		NAB conserved region 1 (NCD1) SAM domain	NAB conserved region 2 (NCD2)	CUX2 (NM_015267) - NAB2 (NM_005967) rearrangement: c.63+83058:CUX2_c.1091+42:NAB2dup Protein fusion: out of frame (CUX2-NAB2)	NA
EVT-P-0038 472-T01-IM 6-13	P-0038 472-T01-IM6		NAB2-DLGAP2	DLGAP2	out-of-frame	five_prime	57487547	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain			NAB2 (NM_005967) - DLGAP2 (NM_004745) rearrangement: t.(8;12)(p23.3;q13.3)(chr8:g.1518542::chr12:g.57487547) Protein Fusion: out of frame {NAB2:DLGAP2}	NA
EVT-P-0039 139-T01-IM 6-14	P-0039 139-T01-IM6		NAB2-KIF5A	KIF5A	in-frame	five_prime	57486074	2	319	True	disrupted	0.718562874251497	True	NAB conserved region 1 (NCD1) SAM domain		NAB conserved region 2 (NCD2)	NAB2 (NM_005967) - KIF5A (NM_004984) rearrangement: c.958-157:NAB2_c.129+5819:KIF5Adeletion Protein Fusion: in frame {NAB2:KIF5A}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022 192-T01-IM 6-18	P-0022 192-T01-IM6		NAB2-LRP1	LRP1	out-of-frame	three_prime	57486554	4	364	True	disrupted	0.017964071856287425	True		NAB conserved region 1 (NCD1) SAM domain	NAB conserved region 2 (NCD2)	LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.842-2141:LRP1_c.1092-146:NAB2dup Protein Fusion: out of frame {LRP1:NAB2}	NA
EVT-P-0028 347-T01-IM 6-21	P-0028 347-T01-IM6		NAB2-MGAT4C	MGAT4C	unknown	five_prime	57486329	3	352	False	disrupted	0.9161676646706587	True	NAB conserved region 1 (NCD1) SAM domain		NAB conserved region 2 (NCD2)	NAB2 (NM_005967) rearrangement: c.1056:NA B2_chr12:g.86871854inv Protein Fusion: mid-exon {NAB2:MGAT4C}	NA
EVT-P-0011 161-T01-IM 5-22	P-0011 161-T01-IM5		NAB2-MGAT4C	MGAT4C	unknown	three_prime	57486534	4	364	True	disrupted	0.017964071856287425	True		NAB conserved region 1 (NCD1) SAM domain	NAB conserved region 2 (NCD2)	NAB2 (NM_005967) rearrangement: c.1092-166:NAB2_chr12:g.87188994del Antisense fusion	NA
EVT-P-0025 245-T01-IM 6-23	P-0025 245-T01-IM6		NAB2-MTOR	MTOR	unknown	three_prime	57491919	7	526	True	lost	0.0	False		NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain		MTOR (NM_004958) - STAT6 (NM_003153) rearrangement: t(1;12)(p36.22;q13.3)(chr1:g.11300310::chr12:g.57491919) Transcript Fusion {MTOR:NAB2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001 055-T02-IM5-49	P-0001 055-T02-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57487938	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon 6 fused with STAT6 exon16) : c.1469-457:NAB2_c.1760:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0066 534-T01-IM7-50	P-0066 534-T01-IM7		NAB2-STAT6	STAT6	unknown	five_prime	57488034	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-361:NAB2_c.1775:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0003 845-T01-IM5-51	P-0003 845-T01-IM5		NAB2-STAT6	STAT6	in-frame	five_prime	57487834	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) -STAT6 (NM_003153) fusion (NAB2 exon 6 fused to STAT6 exon 16) : c.1468+453:NAB2_c.1744+30:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0029 101-T01-IM6-52	P-0029 101-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57486824	4	381	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1144-22:NAB2_c.-1269:STAT6inv Transcript Fusion {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029763-T01-IM6-53	P-0029763-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487587	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1468+206:NAB2_c.1624:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0001055-T03-IM6-55	P-0001055-T03-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487938	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-457:NAB2_c.1760:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0020523-T01-IM6-56	P-0020523-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487505	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused in frame with STAT6 exons 16-22): c.1468+124:NAB2_c.1745-112:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0032069-T01-IM6-57	P-0032069-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487712	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+331:NAB2_c.1870:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022 259-T02-IM 6-61	P-0022 259-T0 2-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57488455	7	510	False	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-7 fused with STAT6 exons 16-22): c.1529:NAB2_c.1745-128:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0004 282-T03-IM 5-62	P-0004 282-T0 3-IM5		NAB2- STAT6	STAT6	in-frame	five_prime	57488115	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exon 16 to 22) : c.1469-280:NAB2_c.1745-157:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0033 539-T02-IM 6-63	P-0033 539-T0 2-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57486902	5	400	False	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) rearrangement: c.1200:NA B2_chr12:g.57486902inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0019 357-T03-IM 6-64	P-0019 357-T0 3-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57488027	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-368:NAB2_c.1690:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019 182-T01-IM 6-66	P-0019 182-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487665	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused in frame with STAT6 exons 16-22): c.1468+284: NAB2_c.1744+39:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0036 288-T01-IM 6-67	P-0036 288-T0 1-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57488128	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1469-267:NAB2_c.1850:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0036 975-T01-IM 6-68	P-0036 975-T0 1-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57486781	4	381	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) rearrangement: c.1143+30: NAB2_chr12:g.57508083inv Transcript Fusion {NAB2:STAT6}	NA
EVT-P-0019 032-T01-IM 6-69	P-0019 032-T0 1-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57488347	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-48:NAB2_c.1691:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0017 518-T01-IM 6-70	P-0017 518-T0 1-IM6		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574865 97	3	364	True	disru pted	0.9880239 520958084	True	NAB c onserv ed region 1 (NC D1) SAM d omain		NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-2 fused to STAT6 exons 18-22): c.1092-103:NAB2_ c.2042:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0016 751-T02-IM 6-71	P-0016 751-T0 2-IM6		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574879 58	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2); NAB c onserv ed region 1 (NC D1) SAM d omain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1469-437:NAB2_ c.1745-109:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0038 591-T01-IM 6-72	P-0038 591-T0 1-IM6		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574877 47	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2); NAB c onserv ed region 1 (NC D1) SAM d omain			NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exon6 fused with STAT6 exon16) : c.14 68+366:NAB2_c.1745-67: STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0016 113-T01-IM 6-73	P-0016 113-T0 1-IM6		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574864 82	3	364	True	disru pted	0.9880239 520958084	True	NAB c onserv ed region 1 (NC D1) SAM d omain		NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon3 fused to STAT6 exon 18: c.1091 +118:NAB2_c.2051:STAT6 inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004 405-T01-IM 5-74	P-0004 405-T0 1-IM5		NAB2- STAT6	STAT6	unknown	five_prime	57487215	6	434	False	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) -STAT6(NM_003153) Fusion (NAB2 exon 6 fused to STAT6 exon 18) : c.1302:NAB2_c.1969:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0004 943-T01-IM 5-75	P-0004 943-T0 1-IM5		NAB2- STAT6	STAT6	unknown	five_prime	57488487	7	521	False	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) reciprocal fusion (NAB2 exons 1-7 with STAT6 exons 15-22) : c.1561:NAB2_c.1739:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0040 364-T01-IM 6-76	P-0040 364-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487738	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:NAB2_c.1892-42:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0040 364-T02-IM 6-77	P-0040 364-T0 2-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487738	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:NAB2_c.1892-42:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014 386-T01-IM 6-78	P-0014 386-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487858	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 17 - 22): c.1468+477:NAB2_c.1891+67:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0043 579-T01-IM 6-79	P-0043 579-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57488047	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-348 :NAB2_c.1744+96:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0044 036-T01-IM 6-80	P-0044 036-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57488037	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-358 :NAB2_c.1744+93:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0044 232-T02-IM 6-81	P-0044 232-T0 2-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57488269	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-126 :NAB2_c.1700:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001 947-T03-IM 6-82	P-0001 947-T03-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487383	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+2: NAB2_c.1769:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0045 590-T01-IM 6-83	P-0045 590-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487780	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399: NAB2_c.1789:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0046 398-T01-IM 6-84	P-0046 398-T01-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487473	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92: NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0046 398-T02-IM 6-85	P-0046 398-T02-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487473	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92: NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner_ gene	fra me_ sta tus	tar get_ role	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0048 656-T01-IM 6-86	P-0048 656-T0 1-IM6		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574864 17	3	364	True	disrup ted	0.9880239 520958084	True	NAB c onserv ed region 1 (NC D1) SAM d omain		NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1091+53: NAB2_c.1956-25:STAT6in v Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0013 489-T01-IM 5-87	P-0013 489-T0 1-IM5		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574875 06	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2); NAB c onserv ed region 1 (NC D1) SAM d omain			NAB2 (NM_005967) - STAT6(NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exons 15 to 22) : c.1468+125:NAB2_ c.1625:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0053 200-T02-IM 6-89	P-0053 200-T0 2-IM6		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574876 09	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2); NAB c onserv ed region 1 (NC D1) SAM d omain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+22 8:NAB2_c.1891+45:STAT6 inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0013 261-T01-IM 5-90	P-0013 261-T0 1-IM5		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574875 67	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2); NAB c onserv ed region 1 (NC D1) SAM d omain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon 6 fused to STAT6 exon 15): c.1468 +186:NAB2_c.1703:STAT6 inv Protein fusion: mid-exon (NAB2-STAT6)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013 216-T02-IM 5-91	P-0013 216-T0 2-IM5		NAB2-STAT6	STAT6	in-frame	five_prime	57487806	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1468+425:NAB2_c.1745-142:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0055 865-T01-IM 6-92	P-0055 865-T0 1-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57488325	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-70: NAB2_c.1744+36:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0055 909-T01-IM 6-93	P-0055 909-T0 1-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57485966	2	319	True	disrupted	0.718562874251497	True	NAB conserved region 1 (NCD1) SAM domain		NAB conserved region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.957+185: NAB2_c.340-42:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0012 841-T01-IM 5-94	P-0012 841-T0 1-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57487686	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 with STAT6 exons 15-22) : c.1468+305:NAB2_c.1712: STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057770-T01-IM6-95	P-0057770-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487467	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+86: NAB2_c.1744+3:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0057786-T01-IM6-96	P-0057786-T01-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57488244	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-151: NAB2_c.1892-23:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0012421-T01-IM5-97	P-0012421-T01-IM5		NAB2-STAT6	STAT6	in-frame	five_prime	57486509	3	364	True	disrupted	0.9880239520958084	True	NAB conserved region 1 (NC D1) SAM domain		NAB conserved region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-3 fused in-frame with STAT6 exons 20-22): c.1091+145: NAB2_c.2159+239:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0007527-T01-IM5-98	P-0007527-T01-IM5		NAB2-STAT6	STAT6	in-frame	five_prime	57488111	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) reciprocal fusion (NAB2 exons 1-6 with STAT6 exons 16-22) : c.1469-284: NAB2_c.1745-161:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001 055-T04-IM6-99	P-0001 055-T04-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57487938	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-457:NAB2_c.1760:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0059 117-T01-IM7-100	P-0059 117-T01-IM7		NAB2-STAT6	STAT6	unknown	five_prime	57488448	7	508	False	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1522:NA B2_c.287:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0059 275-T01-IM7-101	P-0059 275-T01-IM7		NAB2-STAT6	STAT6	unknown	five_prime	57487481	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+100:NAB2_c.1801:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0006 897-T01-IM5-102	P-0006 897-T01-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57488251	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2(NM_005967) - STAT6(NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exons 16 to 22) : c.1469-144:NAB2_c.1811:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA

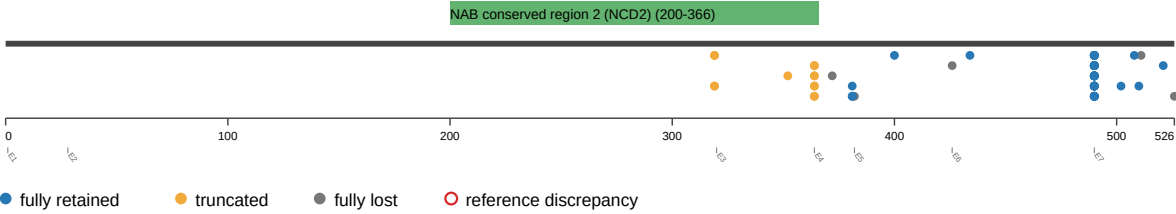
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006 243-T01-IM5-103	P-0006 243-T01-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57488431	7	502	False	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2(NM_005967)-STAT6(NM_003153) Fusion (NAB2 exons 1 to 7 fused with STAT6 exon 16 to 22) : c.1505:NAB2_c.1745-142:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0005 578-T01-IM5-104	P-0005 578-T01-IM5		NAB2-STAT6	STAT6	in-frame	five_prime	57486815	4	381	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exon4 fused with STAT6 exon5) : c.1144-31:NAB2_c.340-81:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0067 841-T01-IM7-106	P-0067 841-T01-IM7		NAB2-STAT6	STAT6	unknown	five_prime	57487573	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+192:NAB2_c.1861:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0066 305-T01-IM7-107	P-0066 305-T01-IM7		NAB2-STAT6	STAT6	in-frame	five_prime	57488138	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2); NAB conserved region 1 (NC D1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-257:NAB2_c.1745-14:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027 038-T01-IM 6-108	P-0027 038-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487452	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2); NAB conserved region 1 (NCD1) SAM domain			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+71:NAB2_c.1744+14:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0030 371-T02-IM 6-109	P-0030 371-T0 2-IM6		NAB2-TMEM117	TMEM117	out-of-frame	five_prime	57485885	2	319	True	disrupted	0.718562874251497	True	NAB conserved region 1 (NCD1) SAM domain		NAB conserved region 2 (NCD2)	NAB2 (NM_005967) - TMEM117 (NM_032256) rearrangement: c.957+104:NAB2_c.278-424:TMEM117dup Protein Fusion: out of frame {NAB2:TMEM117}	NA
EVT-P-0004 638-T01-IM 5-110	P-0004 638-T0 1-IM5		PDZRN4-NAB2	PDZRN4	out-of-frame	three_prime	57486631	4	364	True	disrupted	0.017964071856287425	True		NAB conserved region 1 (NCD1) SAM domain	NAB conserved region 2 (NCD2)	PDZRN4 (NM_001164595) - NAB2 (NM_005967) - rearrangement: c.843+122859:PDZRN4_c.1092-69:NAB2del. Protein fusion: out of frame (PDZRN4-NAB2)	NA
EVT-P-0030 371-T01-IM 6-111	P-0030 371-T0 1-IM6		TMEM117-NAB2	TMEM117	out-of-frame	five_prime	57485885	2	319	True	disrupted	0.718562874251497	True	NAB conserved region 1 (NCD1) SAM domain		NAB conserved region 2 (NCD2)	NAB2 (NM_005967) - TMEM117 (NM_032256) Rearrangement: c.957+104:NAB2_c.278-424:TMEM117dup Protein Fusion: out of frame {NAB2:TMEM117}	NA

Figures

domain retention outliers

NAB2 domain-retention track



fusion schematic

NAB2 fusion-transcript schematic

partner block → breakpoint → retained NAB2 portion (domain-colored, exon ticks)



intragenic deletion schematic

NAB2 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk_impact_50k_2026

